

FOUNDATION BRIDGE

The biology and numbers bridge

What exactly does a biological measurement measure?

By the end, you should be able to...

Distinguish DNA, a gene, RNA and a protein.

Read a small matrix and calculate a fraction.

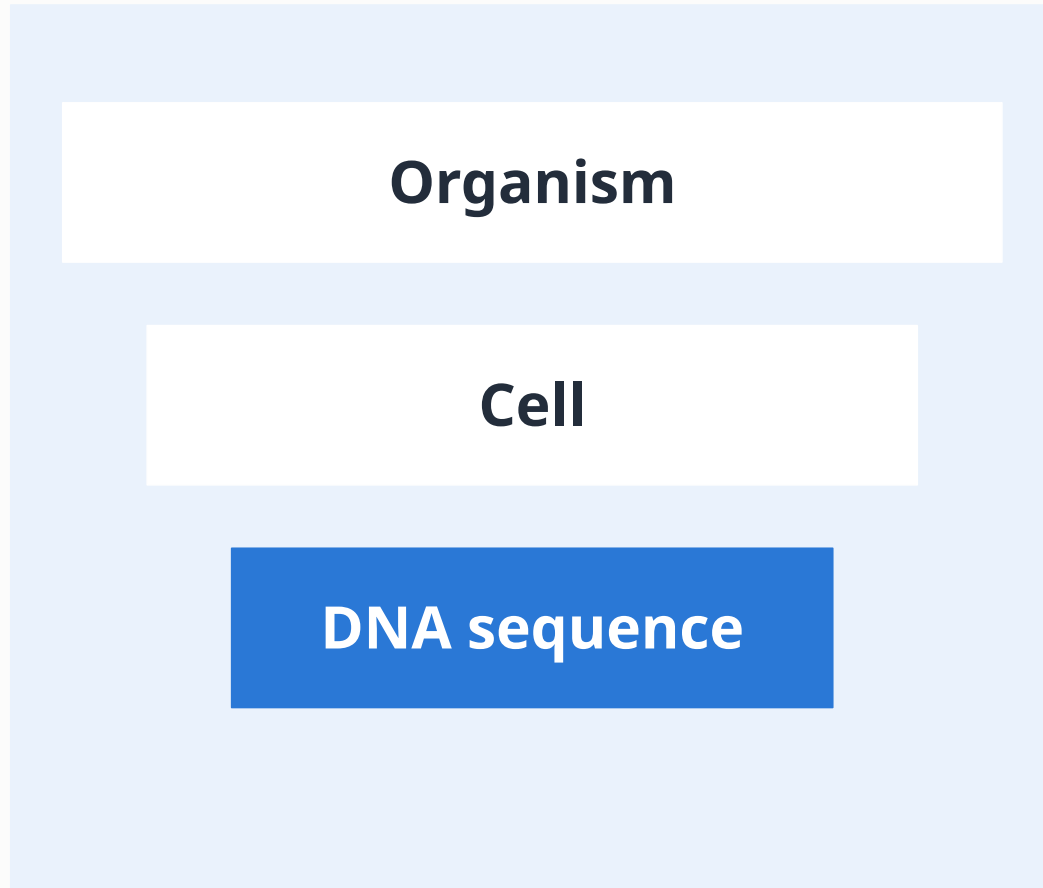
Separate a biological question from the available measurement.

Reconnect with the biology

No previous genetics or programming is required. Start with one cell, one sequence and one table.

Think silently -> explain to a neighbour -> check the measurement.

From organism to sequence



Organism: the whole biological system.

Cell: a compartment containing molecules and processes.

DNA: a sequence that stores heritable information.

A gene is not the same as a protein

A gene is a genomic region associated with a functional product.

Some genes produce functional RNAs, not proteins.

Protein-coding genes can give rise to different transcript isoforms.

The measurement ladder

01

DNA sequence: which bases are present?



02

RNA measurement: which transcripts were detected, and how much?



03

Protein / function assay: which molecule or activity was measured?

Reading a short DNA sequence

5'

3'

ATGCCATN

Direction and alphabet are part of the data.

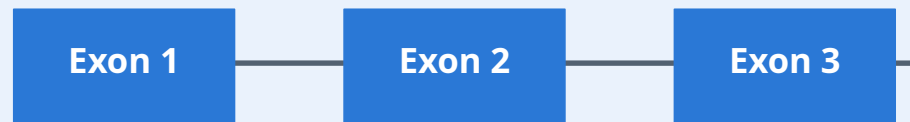
DNA is written with A, C, G and T; RNA uses U instead of T.

Sequences have direction: label 5-prime and 3-prime ends.

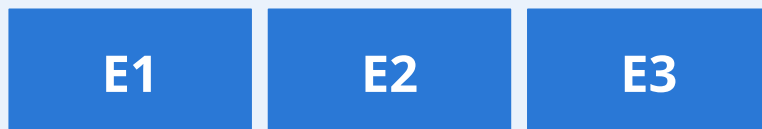
An unknown base N is not a fifth ordinary DNA base.

Exons, introns and transcripts

Precursor RNA



Splicing removes introns



An RNA precursor can include introns that are removed by splicing.

Exons are retained in a mature transcript; not every exon base is translated.

Alternative splicing can produce different transcripts from one gene.

Translation is a reading-frame problem

5'

3'

AUG | GCU | UAA

Met | Ala | Stop

For an appropriate coding sequence, three RNA bases make a codon.

AUG encodes methionine; UAA, UAG and UGA are stop codons in the standard code.

Do not translate an arbitrary genomic interval and assume it is a protein.

Rows, columns and units

Gene	Sample A	Sample B
G1	10	20
G2	30	30
G3	0	4

In this table: rows are genes; columns are samples.

An entry is an observed count, not a number of patients.

Always read the labels before doing arithmetic.

Ratios and logarithms without fear

B / A	Meaning	log2 ratio
2	twice as large	+1
1	unchanged	0
0.5	half as large	-1

A change from 10 to 20 is a two-fold increase.

$\log_2(2) = 1$; $\log_2(1) = 0$; $\log_2(0.5) = -1$.

A denominator of zero needs an explicit rule, not a hidden fix.

Worked example: what can we conclude?

Gene G1 has RNA counts 10 and 20 in two samples.

Predict the result and one limitation before calculating.

Work through the reasoning

The raw count ratio is $20 / 10 = 2$.

We have not checked sequencing depth or biological replication.

The data do not yet establish increased protein abundance or a treatment effect.

Pause: make a defensible claim

A drug-treated sample has more RNA reads for an enzyme. Name one measurement that would test whether enzyme activity also increased.

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

An activity assay with matched controls and biological replicates would address activity. RNA alone is not that assay.

Your preparation route

Use the diagnostic sheet before Week 1. Any missed question directs you to a short bridge section; it does not reduce your grade.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

- 1 gene
- 2 transcript
- 3 protein
- 4 replicate
- 5 ratio

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: draw two different transcripts from the same three-exon gene.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S01 Alberts et al. Molecular Biology of the Cell, 4th ed. Foundational cell biology chapters.

S10 DESeq2: Analyzing RNA-seq data with DESeq2, release vignette.